Sagnik Bhattacharjee

Email | LinkedIn | GitHub

Experience

Research Intern April 2026 – Present

Language Technologies Research Centre, IIIT Hyderabad *Hybrid*

- Building **LensRT**, a computation graph analyzer for the **LiteRT/QAIRT** compile path on **Qualcomm Hexagon**. Maps each NPU partition back to the original ops it absorbed and classifies every rejected op as an *exporter fix*, *SDK limitation*, or *acceptable CPU fallback*.
- Have rewritten unsupported ops in **Conformer** ASR encoders to patterns QAIRT accepts, keeping more of the model on the **Snapdragon NPU** and cutting CPU fallback during inference.

Software Engineer Intern April 2026 - Present

Undisclosed *Remote*

- Built the Capture stage of a teacher-student **distillation pipeline** for **analog circuit optimization**: an LLM agent edits SPICE parameters, runs ngspice, and logs traces that **fine-tune a smaller optimizer model**.
- Cut OPA317 21-parameter convergence from failing to converge in 20 iterations to converging in 4 by adding a **rolling history window** to the optimizer prompt and an **in-memory PDK search tool** over SkyWater SPICE libraries, grounding each move in prior metrics and real W/L binning constraints.
- Added **counterfactual prediction** and **best-of-N comparison capture** to the trace schema, converting single-trajectory logs into **contrastive training pairs** following Process Supervision.
- Built a **cross-file static analyser** for the agents' SPICE netlist pipeline that catches inconsistencies between testbench, topology, and metadata files before they reach the simulator.

Projects

Spheni – Vector Search Library | *C++, Python, OpenMP* | GitHub January 2026

- Architected a vector search library from scratch with **Python bindings (pybind11)**, implementing Flat, IVF-Flat, PQ-Flat, and IVF-PQ indexes with **k-means++ clustering** and **inverted list storage**, achieving **12.6× speedup at 97% Recall@10** on 50k SIFT vectors via Voronoi-partitioned candidate scans.
- Implemented **Product Quantization** with residual encoding and **ADC** via precomputed sub-space lookup tables, compressing a 1M × 768-dim Cohere dataset from **2929 MB to 36.6 MB (80×)** at **83.3% Recall@10-in-100**.
- Eliminated **thread contention** in batch query execution via **thread-local Top-K heaps with OpenMP dynamic scheduling**, scaling throughput from **2.4k to 5.7k QPS (2.37×)** and cutting p99 latency from **0.554 ms to 0.287 ms (1.93×)**.
- Built a **multi-sweep benchmark harness** over 1M vectors characterizing recall–latency–memory tradeoffs across nprobe and M.

sarvam.c – Inference Engine for Sarvam-1 | *C* | GitHub March 2026

- Implemented a **Sarvam-1 inference engine in C** with **OpenBLAS**-accelerated matmul, **GQA**, **RMSNorm**, and **SwiGLU** activations.
- Built a **UTF-8-aware BPE tokenizer** over Sarvam-1's 68k SentencePiece vocabulary with `<0xHH>` byte fallback, greedy merge loop, and `[INST]/[/INST]` chat template injection.
- Implemented a **KV-cache** with per-layer buffers enabling $O(1)$ incremental decoding, with a `RunStats` interface exposing TTFT and throughput metrics.

Skills

Languages: C++, C, Python, SQL
Systems: Linux, OpenMP, Git, Bazel, pybind11
ML Systems: LiteRT, QAIRT, ONNX
Tools: Intel VTune Profiler, perf, GDB, ngspice, OpenBLAS

Education

Indian Institute of Engineering Science and Technology, Shibpur Kolkata, India

BTech in Information Technology *Aug 2025 - July 2029*